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RESEARCH ARTICLE

Forecasting the waste production hierarchical time series with correlation structure

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Abstract

Continuous increase in society's prosperity causes overwhelming growth of the produced municipal solid waste. Circular economy initiatives help to solve this problem by creating closed production cycles, where the produced waste is recycled, or its energy is recovered. An embedment of such principles requires implementation of new waste management strategies. However, these novel strategies must be based on the accurate forecasts of future waste flows. Municipal solid waste production data demonstrate behavior of hierarchical time series. Among all possible approaches to hierarchical times series forecasting, this article is focused on the reconciliation of the base waste generation forecasts. The novel method, that is based on the game-theoretically optimal reconciliation of hierarchical time series, is presented. The modified approach enables to incorporate interdependencies between time series using correlation matrix and to obtain the forecasts corresponding to the unique solution of the optimization problem. The potential of the proposed abstract approach is demonstrated on the waste production data of paper, plastics (both primarily sorted by households), and mixed municipal solid waste from the Czech Republic.

Keywords Prediction · Waste management · Game theory · Zero-sum game · Nash equilibrium · Optimization

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List of symbols

t	Time index
$\boldsymbol{y}_t$	Vector of all observations on all hierarchical levels
$\boldsymbol{S}$	Aggregational consistency summation matrix
$\boldsymbol{b}_t$	Vector of all observations on the lowest level of hierarchy
h	Forecasting horizon
$\widehat{\mathbf{y}}_{\mathrm{t}}(h)$	Base forecasts
$\widetilde{\mathbf{y}}_{\mathrm{t}}(h)$	Reconciled forecasts
$\boldsymbol{G}$	Additional matrix
$\boldsymbol{P}$	Reconciliation matrix
$\boldsymbol{W}_h$	Covariance matrix of the base forecasts' errors
$\boldsymbol{V}_{\boldsymbol{h}}$	Covariance matrix of the reconciled forecasts' errors
$\widehat{\boldsymbol{W}}_1$	Diagonal matrix of the average of base forecasts' squared residuals
$\boldsymbol{Y}$	Vector of the actual future values of time series
$\widetilde{\boldsymbol{Y}}$	Vector of the reconciled aggregate consistent forecasts
$\widehat{\boldsymbol{Y}}$	Vector of the base forecasts
π_1	Payoff function of the forecaster
$\mathcal{A}$	Aggregational consistent forecasts' set
$\mathcal{B}$	Additional constraints' set
π_2	Payoff function of the "randomness"
l	Squared loss function
$\widetilde{\boldsymbol{Y}}_{\mathrm{proj}}$	GTOP optimal forecasts
$a_k, a_{\mathrm{tot}}, a_{kj}, a_{k,\mathrm{tot}}$	Weight coefficients
$\boldsymbol{A}$	Weights matrix
D_F	Function of squared Mahalanobis distance
F	General strictly convex function
$\boldsymbol{x}$	Vector of variables
$\boldsymbol{D}$	Initial weights' matrix
$\widehat{\boldsymbol{R}}$	Estimate of correlation matrix
r_{ij}	Pearson correlation coefficient estimate between time series i and j
$\widehat{e}_{t,i}$	Residual at time t of time series i
$\overline{\widehat{e}}_i$	Average of residuals of time series i
$\mathbf{M}$	Matrix of eigenvalues of $\widehat{\boldsymbol{R}}$
$\boldsymbol{\Lambda}$	Diagonal matrix of eigenvalues of $\widehat{\boldsymbol{R}}$
λ_i	Eigenvalue of $\widehat{\boldsymbol{R}}$
$\widehat{\boldsymbol{R}}_{\mathrm{shrink}}$	Linear shrinkage estimator
v	Shrinkage coefficient
n	Total number of the considered time series
$\boldsymbol{I}_n$	Identity matrix of size n
$\mathrm{var}(r_{ij})$	Estimated variance of r_{ij}
$\boldsymbol{\Lambda}_{\mathrm{shrink}}$	Matrix of shrunken eigenvalues
$\lambda_i^{\mathrm{shrink}}$	Shrunken eigenvalue
n_{eig}	Considered number of eigenvalues

$\boldsymbol{Q}_{\text{spectral}}$	Spectral scaling estimator
$\boldsymbol{W}$	Matrix represented by first n_{eig} columns of $\boldsymbol{M}$
σ^2	Average of the smallest $n - n_{\text{eig}}$ eigenvalues
$\boldsymbol{B}$	Diagonal matrix $\text{diag}(\lambda_1^{\text{shrink}} - \sigma^2, \ldots, \lambda_{n_{eig}}^{\text{shrink}} - \sigma^2)$
$\boldsymbol{\varepsilon}^+, \boldsymbol{\varepsilon}^-$	Multiplicative errors
$\boldsymbol{D}_{\text{mod}}$	Matrix $\boldsymbol{D}$ with entries divided by the corresponding elements of $\widehat{\boldsymbol{Y}}$
$\widetilde{\boldsymbol{\varepsilon}}^+, \widetilde{\boldsymbol{\varepsilon}}^-$	The optimal values of multiplicative errors
SMAPE	Symmetric mean absolute percentage error
F_t	Forecasted values
A_t	Actual future values
m	Number of compared values in SMAPE
p	Suppression parameter

Abbreviations

CE	Circular economy
EU	European union
CEP	Circular economy package
MSW	Municipal solid waste
MSWM	Municipal solid waste management
HTS	Hierarchical time series
MinT	Minimum trace
GTOP	Game-theoretically optimal

1 Introduction

Achieved level of society's development, dwelling in a greater consumption of goods, rapid technological change, and current positive demographic trends lead to a substantial increase in municipal solid waste (MSW) production (Hoornweg et al. 2013). To respond to this tendency many developed countries convert their economy production systems from classical linear models to the system known as circular economy (CE) (Stahel 2016). The main CE principles are minimization of the produced waste and emphasis on the reuse of resources (Velenturf and Purnell 2021).

In the European union (EU), the CE initiatives are legislatively enshrined in the so-called circular economy package (CEP) (Hughes and Purnell 2017). Directives (Directive (EU) 2018/850) and (Directive (EU) 2018/851), being parts of the CEP, set up a series of ambitious milestones for MSW treatment, which must be achieved by the EU countries to successfully embed CE principles. In particular, by the year 2035 only 10% of MSW could be landfilled (Directive (EU) 2018/850), whereas 65% of MSW should be recycled (Directive (EU) 2018/851). These directives are then individually projected to the legislative of each EU state. However, in some states, capacities of currently existing waste recycling or energy recovery facilities are insufficient (Scarlat et al. 2019) and, as a result, their MSW management (MSWM) systems might fail to meet such requirements.

Thus, in order to complete above-mentioned milestones, existing MSWM networks have to be modernized and sufficiently extended. On the other hand, such modernized systems should also be green and financially viable. Therefore, new strategies enabling efficient, economical, and environmentally friendly treatment of MSW should be designed and developed. Such strategies are commonly designed using operation research techniques and optimization models (Ghiani et al. 2014).

However, poor quality input in such models might lead to compromised results, which, for example, might later manifest themselves in building of financially unsustainable oversized MSW treatment facilities (Bisinella et al. 2017). Thus, special attention should be paid to the quality of the input data. Whereas analysis of the current situation plays an important role in MSWM planning and cannot be neglected, long-term strategic decisions should be based on the information about waste composition, production, and sorting level that are expected in the future (Tanskanen et al. 2000). A sustainable MSWM plan must emerge from reliable forecasts considering the necessary time frame, as well as regional and material hierarchies of waste flows. For the purpose of MSW production forecasting, exogenous models and endogenous models or combination of both approaches are often applied (Beigl et al. 2008).

Exogenous models use external information to make prediction or inference about the process. In the case of MSW, most common external factors are social, economic, or demographic variables. Forecasting via an exogenous model is generally a difficult task and requires certain simplifications in predictors, especially when a longer forecasting horizon is considered (due to limited availability of forecasts for predictors). Another possible problem with exogenous models, is that they do not commonly take into account possible changes in the correlation structure, i.e. changes in how predictors affect the forecast (Smejkalová et al. 2020). Nevertheless, exogenous models are among the most popular, since it is easy to ensure cross-comparability of forecasts for such models. For example, in Kaza et al. (2018), MSW forecasts for countries and cities were established using information about expected population and gross domestic product growth. Using the same information accompanied by the additional parameters regarding the plastics fraction, Borrelle et al. (2020) were able to predict plastics pollution of aquatic ecosystems worldwide.

Endogenous analysis of time series helps to avoid such complications, since the only considered explanatory variable is time. Whereas local predictions can produce reliable results for the MSW production in smaller regions (Vu et al. 2019), endogenous prediction of the waste production in a large scale is much more challenging. There are numerous complications that commonly occur during the time series analysis of MSW production, such as shortness of the available data time frame (there is lack of the older data) or inconsistency between regional waste flows on different levels of administrative division.

This paper is devoted to the endogenous approach and is focused on the Czech Republic MSW production forecasting. The waste production annual data (provided by the Ministry of the Environment of the Czech Republic) on the level of microregions for mixed MSW, paper, and plastics are considered. In particular, mixed MSW is waste that has not been yet separated, whereas data on paper and plastics describe

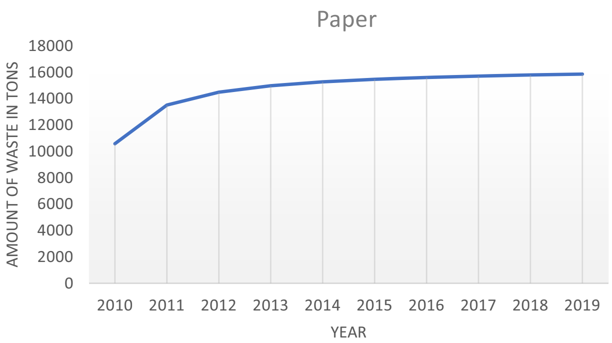

Fig. 1 Time series of paper generation for Ostrava micro-region

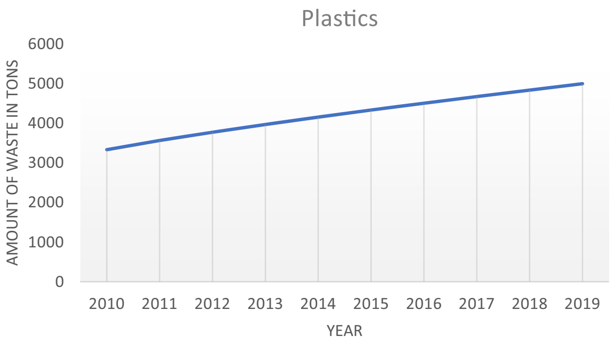

Fig. 2 Time series of plastics generation for Ostrava micro-region

amount of waste that has been primarily sorted by households. The outline of the basic regression model outputs is demonstrated in Figs. 1, 2, 3.

The studied data set exploits behavior of the so-called hierarchical time series (HTS), being a set of time series, which can be structured as a hierarchy with respect to some attribute (Hyndman and Athanasopoulos 2018). In HTS forecasting, historical data are used to design forecasts and a technique is applied on them afterwards in order to achieve aggregational coherency of the forecasts.

However, the main aim of the paper is not to develop new forecasting methodology or to invent definition of weights reflecting reliability of base forecasts. These problems will be solved in accordance with the approach described in (Smejkalová et al. 2022). The main objective of this work is to improve the reconciled HTS MSW production forecasts with the help of additional weights describing interdependencies between time series via correlation estimate.

These weights, reflecting the waste sorting tendencies of households, will be embedded into the minimized objective function describing reconciliation. The approach combines up-to-date theoretical and empirical research in HTS forecasting and has a strong theoretical reasoning. Moreover, the formulation of the problem enables to uniquely determine reconciled forecasts. Although the approach is motivated by the

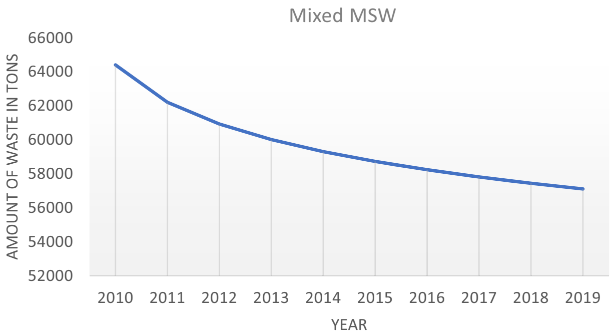

Fig. 3 Time series of mixed MSW generation for Ostrava micro-region

case study of the three above-mentioned waste generation time series from the Czech Republic (through which it will be later verified), the proposed method is abstract and general. Thus, it can be applied to an arbitrary number of distinct MSW fractions, which are assumed to affect each other.

The outline of the article is following. Section 2 will introduce HTS more closely and review approaches that are applied to their forecasting. In particular, basic forecasting methods, such as bottom-up approach, and forecasts reconciliation methods will be reviewed. Section 3 is devoted to a modification of the game-theoretic reconciliation approach, which newly embeds correlation structure into a reconciliation problem. The implications of the underlying metrics choice are thoroughly discussed, demonstrating beneficial theoretical properties of the resulting solution. In Sect. 4, the application of the proposed method to the particular case of the MSWM data of the Czech Republic is presented. Possible modifications of the studied optimization problem and detailed discussion of the obtained results are also given in this section. Section 5 summarizes the main points of the article and concludes it.

2 Hierarchical time series review

Commonly, forecasts of HTS are required to be hierarchically consistent: for example, sum of all forecasts at the lower level of hierarchy are expected to sum up to the forecast for the whole system. From the theoretical point of view, consistency requirement improves overall quality of forecasts (Erven and Cugliari 2015). In practice, the fact that reliability of resulting forecasts increases upwards a hierarchy, was demonstrated on hazardous waste HTS in (Pavlas et al. 2017).

As it was indicated earlier, HTS forecasting is appropriate, when attributes of the observed variable can be viewed as parts of some hierarchical structure and forecasts on different levels of this hierarchy are of interest (Hyndman and Athanasopoulos 2018). For example, the hierarchical division of data can represent the geographical division of the territory (Athanasopoulos et al. 2009). In case of MSW production data, available data can be naturally disaggregated according to the administrative division (state, regions, micro-regions, cities, etc.) or according to the waste fractions (such as recyclable vs. non-recyclable waste, separated vs. non-separated recyclable waste, etc.). Now, the theoretic classification of HTS forecasting will be briefly reviewed.

HTS can be generally classified according to two main attributes: by properties of time aggregation structure or by complexity of aggregation structure, namely, how complicated is mutual interconnection of levels in hierarchical structure, according to which data have to be coherent. Particularly, there are two classes of HTS that differ in properties of time aggregation (Erven and Cugliari 2015):

- Contemporaneous HTS: consistency is required at any given time. Whereas such assumption may seem restrictive, it can be easily fulfilled, when available data are observed with the constant frequency. This is exactly the case of MSW production data, which are usually recorded annually across distinct hierarchical levels.
- Temporal HTS: consistency is required over periods of time during which data are aggregated. For temporal HTS, presence of different time units on different levels is common. For example, when monthly sales of each store in a chain have to correspond to annual sales of the whole chain. Generally, forecasting of temporal HTS is a more complex and challenging task, than forecasting of contemporaneous HTS, since additional summation constraints must be taken in account (Athanasopoulos et al. 2017). In the MSWM field, annual data are the most common, although in some cases, even daily data are available (Song et al. 2014).

According to complexity of aggregation structure HTS can be divided into (Erven and Cugliari 2015):

- HTS: in this case, where is a sole aggregation structure, with respect to which resulting forecasts have to be coherent. In case of MSWM data, work with HTS would mean neglecting either waste composition structure or its regional source and focusing on only one of these attributes.
- Grouped time series: compared to HTS, grouped time series possesses more complicated hierarchical structure, where attributes might "cross" each other or might be nested in one another. For example, MSW production time series, considering not only regional origin of waste, but also particular waste fractions, is a grouped time series.

Regarding the notation, if vector of all observations on all hierarchical levels at time t is denoted as $\boldsymbol{y}_t$ and the vector of all observations on the lowest level of hierarchy at time t is denoted as $\boldsymbol{b}_t$, then hierarchical structure of both above-described aggregation classes can be described using summation matrix $\boldsymbol{S}$ in the following way (Hyndman and Athanasopoulos 2018):

$$\mathbf{y}_{\mathrm{t}} = \mathbf{S}\mathbf{b}_{\mathrm{t}}, \tag{1}$$

To better explain challenges connected with HTS forecasting, short introduction of the commonly applied HTS forecasting methods will be given in the following subsections.

2.1 HTS basic forecasting methods

There are three widely applied basic methods, that avoid reconciliation of the originally obtained forecasts. They are (Hyndman and Athanasopoulos 2018):

- Bottom-up method, which results into coherent forecasts by forecasting of the time series at lowest level of hierarchy and simple summarization of the obtained forecasts upwards according to hierarchical structure. This method is able to produce unbiased forecasts for the whole hierarchy under the assumption of unbiasedness of the bottom level forecasts. Whereas forecasting on the lowest level preserves all available information, it unfortunately causes complications due to noise of time series at the bottom of hierarchy.
- Top-down method, which represents complete opposite to bottom-up method. Comparing to bottom-up approach, here the top level of hierarchy is forecasted and then forecasts are divided downwards according to estimated proportions in aggregational structure. While only one forecasting procedure has to be performed pointing out simplicity of the top-down approaches, there is substantial loss of the information due to aggregation. Moreover, top-down method cannot generate unbiased forecasts and is not suitable for the grouped HTS.
- Middle-out method, which combines two previous methods. Forecasting is performed at some middle level of the hierarchy to which the bottom-up method and top-down method are then applied. Thus, middle-out combines advantages and disadvantages of both methods, while mutually balancing problem with loss of information and noise. Unfortunately, there is no universal rule, that can help to choose the optimal level of forecasting.

2.2 Forecasts reconciliation methods

Forecast reconciliation is based on the idea, that each time series across HTS is forecasted individually and then reconciliation of forecasts is performed to obtain modified coherent forecasts (Hyndman and Athanasopoulos 2018). Assume $\widehat{\mathbf{y}}_t(h)$ denotes base forecasts (above-mentioned individual forecasts) and $\widetilde{\mathbf{y}}_t(h)$ will denote reconciled forecasts, where h is the forecasting horizon. Then, forecast reconciliation method can be represented as

$$\widetilde{\boldsymbol{y}}_{\mathrm{t}}(\mathrm{h}) = \mathbf{SG}\widehat{\boldsymbol{y}}_{\mathrm{t}}(\mathrm{h}) = \mathbf{P}\widehat{\boldsymbol{y}}_{\mathrm{t}}(\mathrm{h}) \tag{2}$$

where $\boldsymbol{P}$ is the reconciliation matrix. All previously described methods can be represented this way, in case of suitable choices of matrix $\boldsymbol{G}$. Thus, they can be seen as particular instances of the reconciliation approach. Using this representation, it can be proven, that top-down method cannot produce unbiased forecasts, since the unbiasedness condition

$$\boldsymbol{SGS} = \boldsymbol{S} \tag{3}$$

cannot be fulfilled for top-down forecasting (Hyndman et al. 2011). Generally, it can be said, that the main aim of the reconciliation approach is to choose matrix $\boldsymbol{P}$ to obtain forecasts that are coherent and have minimal inaccuracies with respect to some metrics. There are two main methods that are used to establish matrix $\boldsymbol{P}$ to obtain accurate forecasts: Minimum Trace (MinT) reconciliation and Game-Theoretically Optimal (GTOP) reconciliation. Firstly, the former approach will be thoroughly reviewed and discussed.

Minimum trace reconciliation. MinT approach focuses on the minimization of forecasts errors with emphasis on error variances under assumption of unbiasedness of the final forecasts (puts the already mentioned $\mathbf{SGS} = \mathbf{S}$ condition on $\boldsymbol{G}$) (Wickramasuriya et al. 2019). Assume the covariance matrix of the base forecasts' errors with time horizon h is denoted as $\boldsymbol{W}_h$, then the covariance matrix of the reconciled forecasts errors is

$$\mathbf{V}_{\mathrm{h}} = \boldsymbol{SG}\mathbf{W}_{\mathrm{h}}\boldsymbol{G}^{\mathrm{T}}\boldsymbol{S}^{\mathrm{T}} \tag{4}$$

The main objective is to minimize the trace of the matrix $\boldsymbol{V}_h$, which determines the sum of error variances. It has been proven, that matrix.

$$\boldsymbol{G} = \left(\boldsymbol{S}^{\mathrm{T}}\mathbf{W}_{\mathrm{h}}^{-1}\mathbf{S}\right)^{-1}\boldsymbol{S}^{\mathrm{T}}\mathbf{W}_{\mathrm{h}}^{-1} \tag{5}$$

will minimize the trace of $\boldsymbol{V}_h$ (Wickramasuriya et al. 2019). Then, $\widetilde{\mathbf{y}}_{\mathbf{t}}(h)$ obtained using.

$$\tilde{\boldsymbol{y}}_{\mathrm{t}}(\mathrm{h}) = \mathbf{S}\left(\boldsymbol{S}^{\mathrm{T}}\mathbf{W}_{\mathrm{h}}^{-1}\mathbf{S}\right)^{-1}\boldsymbol{S}^{\mathrm{T}}\mathbf{W}_{\mathrm{h}}^{-1}\hat{\boldsymbol{y}}_{\mathrm{t}}(\mathrm{h}) \tag{6}$$

is the MinT estimator.

Since computation of $\boldsymbol{W}_h$ (and of its inverse) brings numerous complications, it has to be estimated. As a result, the main MinT volatility dwells in the proposed estimation of $\boldsymbol{W}_h$. For example, the matrix $\boldsymbol{W}_h$ can be replaced by the matrix of weights $\widehat{\boldsymbol{W}}_1$ given by average of base forecasts squared residuals (Hyndman et al. 2016), which is often multiplied by additional proportionality constant. The lists of possible estimators for the can be found in (Athanasopoulos et al. 2017) and in (Nystrup et al. 2020), which extends estimators suggested in the former paper. Whereas such instances of MinT have unique analytical solution, that can be efficiently computed via suitable algorithms (Hyndman et al. 2016), MinT forecasts reconciliation can produce negative values. Such outcome cannot be predicted in advance and is not desirable, when dealing with the HTS, where observed variable cannot be negative (as in case of MSW generation data). Non-negativity improvement of MinT has been suggested in (Wickramasuriya et al. 2020). There, assumption of positive definiteness on $\widehat{\boldsymbol{W}}_1$ has been added to guarantee unique solution of quadratic program, which is the result of non-negativity constraints on the reconciled forecasts.

Game-theoretically optimal reconciliation. GTOP method emerged as an alternative to all above-mentioned methods. The previously presented approaches do not distinct between purposes of aggregate consistency and hierarchical information sharing (Erven and Cugliari 2015), and, as a result, might have restrictive theoretical requirements. GTOP eliminates this problem by dividing the forecasting into two independent steps: it solely focuses on the aggregate consistency achieved via Nash Equilibrium, and ignores interdependencies of the hierarchical levels, leaving this as an open problem (Erven and Cugliari 2015). Thus, GTOP can be applied to every HTS universally, but does not take possible correlations into account. This approach will be discussed in Methods section in detail.

2.3 Novelty and contribution

Whereas none of these methods can be definitively considered as the best, the nature of the studied data itself can narrow spectrum of approaches, that can be applied. The MSWM data available for this research are considered to be contemporaneous HTS. In case unbiasedness of the forecasts is not priority, the top-down method can be of interest. It can be realized through waste generation disaggregation approach presented in (Madden et al. 2021). When unbiasedness of final forecasts is required, only the bottom-up method can be applied to unbiasedly forecast considered HTS. However, since there might be numerous time series at the bottom level of the hierarchy, great potential of noise is expected. Existence of this problem for the MSWM data has been pointed out in (Rosecký et al. 2021). Moreover, these methods do not consider the inherent hierarchical correlation structure (Hyndman and Athanasopoulos 2018). Whereas, previous MinT instances resulted into analytical solution, which has been pointed out as a substantial advantage compared to GTOP algorithmic solution, non-negative MinT loses this advantage. The GTOP can be considered as a suitable algorithm for the MSWM data, since it has no restrictions. The GTOP MSW production prediction is expected to be a solution to the quadratic program which can be handled by an off-the-shelf optimization software. However, as it was already mentioned, the GTOP approach still considers only aggregate consistency, and the information sharing should be considered separately. The embedding of information sharing throughout hierarchy in the GTOP algorithm would be the main goal of this paper, which is expected to provide better forecasts of the future waste flows. Now, the contribution of this work to the currently existing waste forecasting methodologies in particular will be discussed.

In the well-established seminal reports such as Kaza et al. (2018) and Borrelle et al. (2020), which have been already mentioned in Introduction section, the exogenous models have been studied. Thus, both models bear drawbacks of exogenous models and the approach presented in this paper can serve as a complement to them. Also, the above-mentioned studies consider counties as a whole, or only some larger cities. Therefore, there is no requirement for the coherency of the resulting forecasts, whereas the available predictors' forecasts are already coherent in case the forecasting is performed on smaller grids (Lebreton and Andrady 2019). Moreover, in those reports, waste fractions/types such as plastics are considered separately (Borrelle et al. 2020) or MSW is studied as a whole (Kaza et al. 2018), thus these models are not able to embed dependencies between different MSW fractions and project them onto forecasts: uncertanties in changes of proportion of plastics in waste are either represented using some estimated interval from the data (Lebreton and Andrady 2019) or using Monte-Carlo simulation (Borrelle et al. 2020).

Thus, in this article, contemporaneous endogenous HTS forecasting of annual amount plastics, paper, and mixed MSW according to administrative division will be studied. Forecasts of each hierarchy level itself and calculation of accuracy weights will be omitted since their fully copy an approach already presented in (Smejkalová et al. 2022). The reconciled forecasts will be achieved through the solution of the quadratic mathematical program and their theoretical properties will be discussed. The method will not only ensure coherency between forecasts on different hierarchy levels, but the result will also take into account how the considered waste types affect each other. Thus, this result "extrapolates" the waste separation dependencies previously described in (Šomplák et al. 2022). Moreover, using the analysis of the initial results, the robust modification of the studied optimization problem for the corrupted inputs/weights will be suggested. The contribution of this paper can be summarized as follows:

- The original optimization problem will enable to consider interdependencies between waste fractions, reflecting waste sorting trends present in the data, which is crucial for MSWM forecasting.
- The introduced forecast loss term will be based on Bregman divergence (Cesa-Bianchi and Lugosi 2009) and, consequently, the optimally reconciled forecasts will uniquely exist.
- The GTOP approach will enable to account for non-negativity of forecasts straightforwardly.

3 Methods

As well as the previously mentioned HTS forecasting methods (Hyndman and Athanasopoulos 2018), GTOP method can also be viewed as mapping from the set of base forecasts to the set of aggregate consistent final forecasts. GTOP considers forecasts reconciliation to be a two-players zero-sum game. It is a normal-form game with two players, where profit of the one player always equals to the loss of the other (Owen 1995). First player is a "forecaster", who tries to minimize his payoff function

$$\pi_1(\mathbf{Y}, \widetilde{\mathbf{Y}}) = \mathrm{l}(\mathbf{Y}, \widetilde{\mathbf{Y}}) - \mathrm{l}(\mathbf{Y}, \widehat{\mathbf{Y}}) \tag{7}$$

which represents difference between a squared loss of the reconciled aggregate consistent forecasts $\widetilde{\boldsymbol{Y}} = (\widetilde{Y}_1, \ldots, \widetilde{Y}_k, \ldots, \widetilde{Y}_{\mathrm{tot}})$

$$\mathrm{l}\left(\mathbf{Y}, \tilde{\mathbf{Y}}\right) = \sum_{k} \mathrm{l}_k\left(\mathrm{Y}_k, \tilde{\mathrm{Y}}_k\right) + \mathrm{l}_{\mathrm{tot}}\left(\mathrm{Y}_{\mathrm{tot}}, \tilde{\mathrm{Y}}_{\mathrm{tot}}\right) = \sum_{k} \mathrm{a}_k\left(\mathrm{Y}_k - \tilde{\mathrm{Y}}_k\right)^2 + \mathrm{a}_{\mathrm{tot}}\left(\mathrm{Y}_{\mathrm{tot}} - \tilde{\mathrm{Y}}_{\mathrm{tot}}\right)^2 \tag{8}$$

where $\boldsymbol{Y} = (Y_1, \ldots, Y_k, \ldots, Y_{\text{tot}})$ is a vector of an actual future values of time series (both vectors represent the same time moment), and a squared loss of the base forecasts $l(\boldsymbol{Y}, \widehat{\mathbf{Y}})$ defined in a same way with $\widehat{\boldsymbol{Y}} = (\widehat{Y}_1, \ldots, \widehat{Y}_k, \ldots, \widehat{Y}_{\text{tot}})$ representing a vector of base forecasts. Individual positive weights of time series reflecting noise and prediction errors are embedded into the squared loss using coefficients $a_k > 0$, $a_{\text{tot}} > 0$. The "forecaster's" strategy set $\mathcal{A}$ represents all possible aggregate consistent forecasts $\widetilde{\boldsymbol{Y}} \in \mathcal{A}$. Therefore, "forecaster" seeks smaller level of squared loss for reconciled forecasts to outperform the base forecasts. Another player is an "uncertainty", who plays against the "forecaster" and tries to minimize the "improvement" that "forecaster" can possibly achieve. Thus,

$$\pi_2(\mathbf{Y}, \widetilde{\mathbf{Y}}) = -\pi_1(\mathbf{Y}, \widetilde{\mathbf{Y}}) \tag{9}$$

The "uncertainty" plays with the set $\mathcal{A} \cap \mathcal{B}$ of aggregate consistent HTS data $\mathcal{A}$ and further possible constraints $\mathcal{B}$ with an element $\boldsymbol{Y} \in \mathcal{A} \cap \mathcal{B}$.

The minimax value of the "forecaster's" payoff function is of interest: it is the smallest value (the maximal improvement) that the "forecaster" can obtain no matter what "uncertainty" plays. Exactly, the strategy played by the "forecaster" in the minimax value is the GTOP reconciled forecasts. The minimax value can be represented as

$$\min_{\boldsymbol{Y} \in \mathcal{A} \cap \mathcal{B}} \max_{\tilde{\boldsymbol{Y}} \in \mathcal{A}} l\left(\boldsymbol{Y}, \tilde{\boldsymbol{Y}}\right) - l\left(\boldsymbol{Y}, \hat{\boldsymbol{Y}}\right) \tag{10}$$

and it has been shown in (Erven and Cugliari 2015), that the Nash equilibrium (Owen 1995) of this game and the resulting GTOP prediction can be computed as follows

$$\widetilde{\mathbf{Y}}_{\text{proj}} = \arg \min_{Y \in A \cap B} \mathbf{l}(\widehat{\boldsymbol{Y}}, \widetilde{\boldsymbol{Y}}) \tag{11}$$

Optimal reconciliation of forecasts is denoted as $\widetilde{\boldsymbol{Y}}_{\text{proj}}$ since it has been proven that such reconciliation is in fact orthogonal projection of base forecasts onto set of reconciled forecasts in a geometric sense (Panagiotelis et al. 2021). Under assumption, that $\mathcal{B}$ is closed and convex set, such GTOP prediction uniquely exists. Set $\mathcal{A}$ just has to be non-empty, implying meaningfulness of the considered hierarchical structure. In case both sets can be described as a finite set of linear constraints, GTOP solution can be found using quadratic programming techniques. Clearly, such an approach can be extended to the whole forecasting horizon by solving this problem at each considered time moment individually.

In this article, implementation of the information sharing into the GTOP is suggested through utilization of another loss measure instead of the one previously presented. The newly applied loss has to be defined in a such way, that will penalize not only losses between data and estimates but also unreasoned differences between correlated estimates. This way it will take into account possible interdependencies over different administrative levels and waste types. Therefore, suggested definition of loss is following.

$$\mathrm{l_{kj}}\left(\mathrm{Y_k}, \hat{\mathrm{Y}}_\mathrm{k}, \mathrm{Y_j}, \hat{\mathrm{Y}}_\mathrm{j}\right) = \mathrm{a_{kj}}\left(Y_k - \hat{\mathrm{Y}}_\mathrm{k}\right)\left(Y_j - \hat{\mathrm{Y}}_\mathrm{j}\right), \tag{12}$$

$$\mathrm{l_{k,tot}}\left(\mathrm{Y_k}, \widehat{\mathrm{Y}}_\mathrm{k}, \mathrm{Y_{tot}}, \widehat{\mathrm{Y}}_\mathrm{tot}\right) = \mathrm{a_{k,tot}}\left(Y_k - \widehat{\mathrm{Y}}_\mathrm{k}\right)\left(\mathrm{Y_{tot}} - \widehat{\mathrm{Y}}_\mathrm{tot}\right) \tag{13}$$

and analogically defined $l_{\mathrm{tot},k}\left(Y_{\mathrm{tot}}, \widehat{Y}_{\mathrm{tot}}, Y_k, \widehat{Y}_k\right)$ and $l_{\mathrm{tot,tot}}\left(Y_{\mathrm{tot}}, \widehat{Y}_{\mathrm{tot}}, Y_{\mathrm{tot}}, \widehat{Y}_{\mathrm{tot}}\right)$. Then, the new total squared loss, is defined as

$$\begin{aligned}\mathrm{l}\left(\mathbf{Y}, \widehat{\mathbf{Y}}\right) = &\sum_{\mathrm{k,j}} \mathrm{l_{kj}}\left(\mathrm{Y_k}, \widehat{\mathrm{Y}}_\mathrm{k}, \mathrm{Y_j}, \widehat{\mathrm{Y}}_\mathrm{j}\right) + \sum_{\mathrm{k}} \mathrm{l_{k,tot}}\left(\mathrm{Y_k}, \widehat{\mathrm{Y}}_\mathrm{k}, \mathrm{Y_{tot}}, \widehat{\mathrm{Y}}_\mathrm{tot}\right)\\ &+ \sum_{\mathrm{k}} \mathrm{l_{tot,\,k}}\left(\mathrm{Y_{tot}}, \widehat{\mathrm{Y}}_\mathrm{tot}, \mathrm{Y_k}, \widehat{\mathrm{Y}}_\mathrm{k}\right) + \mathrm{l_{tot,\,tot}}\left(\mathrm{Y_{tot}}, \widehat{\mathrm{Y}}_\mathrm{tot}, \mathrm{Y_{tot}}, \widehat{\mathrm{Y}}_\mathrm{tot}\right)\end{aligned} \tag{14}$$

Clearly, the $l\left(\boldsymbol{Y}, \widetilde{\boldsymbol{Y}}\right)$ is defined in an analogical way. This novel squared loss can also be represented using matrix–vector notation. Then,

$$\mathrm{l}\left(\mathbf{Y}, \widetilde{\mathbf{Y}}\right) = \left(\mathbf{Y} - \widetilde{\mathbf{Y}}\right)^{\mathbf{T}}\mathbf{A}\left(\mathbf{Y} - \widetilde{\mathbf{Y}}\right) \tag{15}$$

and

$$\mathrm{l}\left(\mathbf{Y}, \widehat{\mathbf{Y}}\right) = \left(\mathbf{Y} - \widehat{\mathbf{Y}}\right)^{\mathbf{T}}\mathbf{A}\left(\mathbf{Y} - \widehat{\mathbf{Y}}\right) \tag{16}$$

Now, the theoretical properties of this loss have to be studied in detail in order to make conclusions about uniqueness and existence of the game-theoretically optimal prediction based on this loss. (Erven and Cugliari 2015) have proven that the existence and uniqueness of the GTOP prediction holds for every loss function based on a Bregman divergence, due do the fact that the generalized Pythagorean inequality holds for this type of divergence function (Cesa-Bianchi ang Lugosi, 2009). Thus, to obtain the analogical results, it is sufficient to prove, that newly defined loss term is Bregman divergence. Currently defined loss is a particular instance of the squared Mahalanobis distance (De Maesschalck et al. 2000), which can be seen as a generalized instance of the Euclidean distance applied in (Erven and Cugliari 2015). The squared Mahalanobis distance

$$\mathrm{D_F}(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^{\mathrm{T}}\mathbf{A}(\mathbf{x} - \mathbf{y}) \tag{17}$$

is a Bregman divergence under assumption that it is generated by the strictly convex function

$$F(\mathbf{x}) = \frac{1}{2}\mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x} \tag{18}$$

defined on a closed convex set (Fischer 2010) (holds for the assumed $\mathcal{A} \cap \mathcal{B}$). The strict convexity of the $\mathrm{F}(\boldsymbol{x})$ is guaranteed for the positive definite matrix $\boldsymbol{A}$ (Boyd 2004).

Thus, in order to reproduce existence and uniqueness of the GTOP prediction for the new loss term, it is crucial to choose coefficients in a matrix $\boldsymbol{A}$ in a such way, which will guarantee the positive definiteness of the matrix and will reasonably reflect the interdependencies inside the hierarchical structure.

To embed the interdependencies into the loss term, the following matrix

$$\mathbf{A} = \mathbf{D}\widehat{\mathbf{R}}^{-1}\mathbf{D} \tag{19}$$

is proposed. A matrix $\boldsymbol{D}$ plays the same role as it was in the originally presented GTOP: it is the diagonal matrix of positive weights assigned to each observed time series that reflects the accuracy/reliability of the base forecasts. The matrix $\widehat{\boldsymbol{R}}$ is assumed to be an estimate of correlation matrix of the base forecasts' residuals. It is known that the correlation matrix is only positive semi-definite (Johnston 2021), which is not enough, since it will fail to fulfill strictness of convexity and as result suggested loss will not be based on the Bregman divergence. Clearly, due to strong aggregational interdependencies and, possibly, the fact that number of time series is greater than number of observations, the matrix $\widehat{\boldsymbol{R}}$ will be ill-conditioned and rank-deficient. As a result, even calculation of Moore–Penrose pseudo-inverse will produce numerically unstable $\widehat{\boldsymbol{R}}^{-1}$. To overcome this obstacle, we employ spectral scaling estimator $\boldsymbol{Q}_{\text{spectral}}$ (Nystrup et al. 2021) instead of $\widehat{\boldsymbol{R}}^{-1}$. The empirical correlation matrix will be denoted as $\widehat{\boldsymbol{R}} = \left[r_{ij}\right]$, where

$$\mathrm{r_{ij}} = \frac{\sum_{\mathrm{t}}\left(\widehat{\mathrm{e}}_{\mathrm{t,i}} - \bar{\widehat{\mathrm{e}}}_{\mathrm{i}}\right)\left(\widehat{\mathrm{e}}_{\mathrm{t,j}} - \bar{\widehat{\mathrm{e}}}_{\mathrm{j}}\right)}{\sqrt{\sum_{\mathrm{t}}\left(\widehat{\mathrm{e}}_{\mathrm{t,i}} - \bar{\widehat{\mathrm{e}}}_{\mathrm{i}}\right)^2}\sqrt{\sum_{\mathrm{t}}\left(\widehat{\mathrm{e}}_{\mathrm{t,j}} - \bar{\widehat{\mathrm{e}}}_{\mathrm{j}}\right)^2}} \tag{20}$$

is an estimate of correlation coefficient between base forecast errors with $i, j \in (1, \ldots, k, \ldots, \text{tot})$. Such simple correlation estimate can be considered as suitable since the considered time series are represented only by a trend, and presence of some autocorrelation can be hardly assumed. This estimate can be decomposed as

$$\widehat{\mathbf{R}} = \mathbf{M}\boldsymbol{\Lambda}\mathbf{M}^{\mathrm{T}} \tag{21}$$

such that $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_n)$, where $\lambda_1, \ldots, \lambda_\mathrm{n}$ are eigenvalues of $\widehat{\boldsymbol{R}}$ and $\boldsymbol{M}$ is a matrix of eigenvectors, with columns ordered in descending order in accordance with value of corresponding eigenvalue. This matrix can be modified by a linear shrinkage estimator.

$$\widehat{\mathbf{R}}_{\text{shrink}} = (1 - \mathrm{v})\widehat{\mathbf{R}} + \mathrm{v}\mathbf{I}_\mathrm{n} \tag{22}$$

which is positive-definite and well-conditioned for $v > 0$. In accordance with (Nystrup et al. 2021), shrinkage parameter will be set as

$$\mathrm{v} = \frac{\sum_{\mathrm{i}\neq\mathrm{j}}\mathrm{var}\left(\mathrm{r_{ij}}\right)}{\sum_{\mathrm{i}\neq\mathrm{j}}\mathrm{r_{ij}^2}} \tag{23}$$

where variance of correlation estimate will be calculated as

$$\mathrm{var}\left(\mathrm{r_{ij}}\right) = \frac{\left(1 - \mathrm{r}_{\mathrm{ij}}^{2}\right)^{2}}{\mathrm{n}} \tag{24}$$

The $\widehat{\boldsymbol{R}}_{\mathrm{shrink}}$ can be also rewritten as

$$\widehat{\mathbf{R}}_{\mathrm{shrink}} = \mathbf{M}\Lambda_{\mathrm{shrink}}\mathbf{M}^{\mathrm{T}} \tag{25}$$

where $\boldsymbol{\Lambda}_{\mathrm{shrink}} = \mathrm{diag}\left(\lambda_1^{\mathrm{shrink}}, \ldots, \lambda_n^{\mathrm{shrink}}\right)$ and $\lambda_i^{\mathrm{shrink}} = (1 - v)\lambda_i + v$. Then, well-conditioned and positive-definite $\boldsymbol{Q}_{\mathrm{spectral}}$ can be obtained using first $n_{\mathrm{eig}} \leq n$ eigenvalues of $\widehat{\boldsymbol{R}}$ as

$$\mathbf{Q}_{\mathrm{spectral}} = \left(\mathbf{W}\mathbf{B}\mathbf{W}^{\mathrm{T}} + \boldsymbol{\sigma}^{2}\mathbf{I}_{\mathbf{n}}\right)^{-1} \tag{26}$$

where $\boldsymbol{W}$ is an $n \times n_{\mathrm{eig}}$ matrix represented by first n_{eig} columns of $\boldsymbol{M}$,

$$\mathbf{B} = \mathrm{diag}(\lambda_1^{\mathrm{shrink}} - \sigma^2, \ldots, \lambda_{\mathrm{n_{eig}}}^{\mathrm{shrink}} - \sigma^2) \tag{27}$$

and

$$\sigma^2 = \frac{\sum_{\mathrm{j=n_{eig}+1}}^{\mathrm{n}} \lambda_{\mathrm{j}}^{\mathrm{shrink}}}{\mathrm{n} - \mathrm{n_{eig}} - 1} \tag{28}$$

Thus, the resulting matrix

$$\mathbf{A} = \mathbf{D}\mathbf{Q}_{\mathrm{spectral}}\mathbf{D} = \mathbf{D}\left(\mathbf{W}\mathbf{B}\mathbf{W}^{\mathrm{T}} + \sigma^{2}\mathbf{I}_{\mathrm{n}}\right)^{-1}\mathbf{D} \tag{29}$$

is assumed to be a product of diagonal matrix with positive entries (therefore, it is positive definite (Johnston 2021), estimate of inverse of a correlation matrix (which is also positive definite by construction) and again diagonal matrix with positive entries. Under these assumptions the symmetric matrix $\boldsymbol{A}$ will be positive-definite (Johnston 2021) and will guarantee that the suggested type of loss corresponds to Bregman divergence of the Legendre function if the underlying function domain is convex. Therefore, even for this newly defined type of loss, the saddle point of the loss minimization game or argument of the solution of the corresponding quadratic program will always exist, will be unique and will outperform any other possible aggregate consistent reconciled forecast in terms of chosen metrics. The resulting optimal reconciled forecast is then

$$\widetilde{\mathbf{Y}}_{\mathrm{proj}} = \arg \min_{\widetilde{Y} \in A \cap B} \left(\widehat{\mathbf{Y}} - \widetilde{\mathbf{Y}}\right)^{\mathrm{T}}\mathbf{A}\left(\widehat{\mathbf{Y}} - \widetilde{\mathbf{Y}}\right) \tag{30}$$

where convex set $\mathcal{B}$ represents additional non-negativity assumption. It is important to highlight, that the proposed GTOP-based approach has led to the mathematical program with the objective function analogical to the one minimized by MinT predictions (Wickramasuriya et al. 2020). This fact emphasizes the evident connection between GTOP and MinT estimators previously pointed out in the already mentioned (Panagiotelis et al. 2021). Whereas in (Wickramasuriya et al. 2020), the optimization problem has been obtained through the modification of the MinT approach, in this article, the analogical results have been obtained via theoretical background of game theory solution with Bregman divergence defined as Mahalanobis distance.

4 Results and discussion

The proposed approach has been applied to MSWM data from the Czech Republic. The available time series describe annual production of plastic, paper and mixed MSW on the levels of micro-regions from year 2010 till year 2021 (observations for regions and the national level have been calculated according to the hierarchical summation matrix). As it has been already mentioned, the study is aimed at demonstration of the fact that the proposed modification of the GTOP approach is able to provide better predictions based on the already available diagonal weights matrices $\boldsymbol{D}$ obtained according to (Smejkalová et al. 2022). It is assumed, that incorporation of the interdependencies within the considered MSWM HTS will enable to better describe households' waste sorting tendencies.

Using the non-negative error variables $\boldsymbol{\varepsilon}^+$ and $\boldsymbol{\varepsilon}^-$, such that

$$\left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}}=\widetilde{\mathbf{Y}} \tag{31}$$

instead of original errors $\left(\widehat{Y}-\widetilde{Y}\right)$ from (30), the problem

$$\min_{\widetilde{Y}\in A\cap B}\left(\widehat{\mathbf{Y}}-\widetilde{\mathbf{Y}}\right)^{\mathrm{T}}\mathbf{D}_{\mathrm{mod}}^{2}\left(\widehat{\mathbf{Y}}-\widetilde{\mathbf{Y}}\right) \tag{32}$$

with the choice

$$\mathbf{D}_{\mathrm{mod}}=\operatorname{diag}\left(\frac{\mathrm{d}_1}{\widehat{\mathrm{Y}}_1},\ldots,\frac{\mathrm{d}_{\mathrm{tot}}}{\widehat{\mathrm{Y}}_{\mathrm{tot}}}\right),\ \text{with}\ \mathrm{d}_{\mathrm{i}}=\mathrm{d}_{\mathrm{ii}}\ \text{of}\ \mathbf{D} \tag{33}$$

can be rewritten as

$$\min_{\boldsymbol{\varepsilon}^{+},\boldsymbol{\varepsilon}^{-}}\left(\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)^{\mathrm{T}}\mathbf{D}^{2}\left(\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right) \tag{34}$$

$$\text{s.t.}\ \left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}}\geq 0 \tag{35}$$

$$\left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}}=\mathbf{S}\left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}} \tag{36}$$

$$\boldsymbol{\varepsilon}^{+}\geq 0 \tag{37}$$

$$\boldsymbol{\varepsilon}^- \geq 0 \tag{38}$$

where $\boldsymbol{S}$ is the corresponding hierarchy summation matrix. Then, the projection of base forecasts onto reconciled non-negative forecasts is

$$\widetilde{\mathbf{Y}}_{\mathrm{proj}} = \widehat{\mathbf{Y}}(1 + \widetilde{\boldsymbol{\varepsilon}}^+ - \widetilde{\boldsymbol{\varepsilon}}^-), \tag{39}$$

with

$$\widetilde{\boldsymbol{\varepsilon}}^+, \widetilde{\boldsymbol{\varepsilon}}^- \in \arg \min_{\boldsymbol{\varepsilon}^+, \boldsymbol{\varepsilon}^-} \left(\boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right)^{\mathrm{T}} \mathbf{D}^2 \left(\boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right) \tag{40}$$

The optimization problem (34–38) is analogical to the one that is presented in (Smejkalová et al. 2022). Thus, the study of (Smejkalová et al. 2022) is a minimization of the proposed Bregman divergence with the particular choice of matrix $\boldsymbol{D}^2_{\boldsymbol{mod}}$. Therefore, in order to stay in premise with the aim of the paper, the problem

$$\min_{\boldsymbol{\varepsilon}^+, \boldsymbol{\varepsilon}^-} \left(\boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right)^{\mathrm{T}} \mathbf{D}\mathbf{Q}_{\mathrm{spectral}}\mathbf{D}\left(\boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right) \tag{41}$$

$$\text{s.t. } \left(1 + \boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right)\widehat{\mathbf{Y}} \geq 0 \tag{42}$$

$$\left(1 + \boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right)\widehat{\mathbf{Y}} = \mathbf{S}\left(1 + \boldsymbol{\varepsilon}^+ - \boldsymbol{\varepsilon}^-\right)\widehat{\mathbf{Y}} \tag{43}$$

$$\boldsymbol{\varepsilon}^+ \geq 0 \tag{44}$$

$$\boldsymbol{\varepsilon}^- \geq 0 \tag{45}$$

will be solved. To assess the resulting GTOP predictions accuracy, it has been suggested to remove the last 2 years (2020 and 2021) and produce base forecasts for years 2010–2021 based on the data from 2010–2019 (this time frame was also used to calculate $\boldsymbol{Q}_{\mathrm{spectral}}$ with the n_{eig} corresponding to the rank of $\widehat{\boldsymbol{R}}$). Then, the reconciliation of the base forecasts for years 2020 and 2021 has been performed. The SMAPE

$$\mathrm{SMAPE} = \frac{1}{\mathrm{m}} \sum_{\mathrm{i}=1}^{\mathrm{m}} \frac{|\mathrm{F_i} - \mathrm{A_i}|}{(\mathrm{A_i} + \mathrm{F_i})/2} \tag{46}$$

where F_i is the forecast value, A_i is an actual future value, and m is the total number of the compared values, has been chosen as an accuracy measure and has been computed for each administrative unit and each waste type. Table 1 provides a comparison of the reconciliation based only on the initial weights and the approach proposed in this paper.

Unfortunately, the obtained predictions do not outperform the predictions made without consideration of the correlation structure. Possible reason of the resulting

Table 1 Comparison of initial and new forecasts SMAPE values [%]

Method\Waste type	Paper	Plastics	Mixed MSW	Total
Without correlation	17.603	8.331	5.184	10.373
With correlation	18.139	9.530	6.467	11.379

Table 2 SMAPE values of the approach without opposite correlations [%]

Method\Waste type	Paper	Plastics	Mixed MSW	Total
Without terms $\boldsymbol{\varepsilon}^{-}\boldsymbol{D}\boldsymbol{Q}_{\text{spectral}}\boldsymbol{D}\boldsymbol{\varepsilon}^{+}$	17.414	8.347	5.181	10.314

inaccuracies might be the fact, that individual weights of time series are suppressed by an interdependencies between noised time series. Therefore, quality of the forecasts itself is ignored in favor of mutual compliance. A more detailed analysis has implied that the main source of the observed distortion compared to the approach without correlation are the terms of type

$$\boldsymbol{\varepsilon}^{-}\mathbf{D}\mathbf{Q}_{\text{spectral}}\mathbf{D}\boldsymbol{\varepsilon}^{+} \tag{47}$$

Indeed, if the terms (47) are removed from the minimization problem, then the SMAPE values demonstrate some improvement as presented in Table 2.

Then, the following modification that suppresses impact of these terms has been proposed

$$\min_{\boldsymbol{\varepsilon}^{+},\boldsymbol{\varepsilon}^{-}} \mathrm{p}\left(\boldsymbol{\varepsilon}^{+}+\boldsymbol{\varepsilon}^{-}\right)^{\mathrm{T}}\mathbf{D}\mathbf{Q}_{\text{spectral}}\mathbf{D}\left(\boldsymbol{\varepsilon}^{+}+\boldsymbol{\varepsilon}^{-}\right)+(1-\mathrm{p})\left(\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)^{\mathrm{T}}\mathbf{D}\mathbf{Q}_{\text{spectral}}\mathbf{D}\left(\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right) \tag{48}$$

$$\text{s.t. } \left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\hat{\mathbf{Y}} \geq 0 \tag{49}$$

$$\left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}} = \mathbf{S}\left(1+\boldsymbol{\varepsilon}^{+}-\boldsymbol{\varepsilon}^{-}\right)\widehat{\mathbf{Y}} \tag{50}$$

$$\boldsymbol{\varepsilon}^{+} \geq 0 \tag{51}$$

$$\boldsymbol{\varepsilon}^{-} \geq 0 \tag{52}$$

Clearly, such modification does not preserve initial GTOP properties of the forecast, but still has a unique solution due do positive definiteness of $\boldsymbol{D}\boldsymbol{Q}_{\text{spectral}}\boldsymbol{D}$. The performance of the modification for $p = 0.2$ (the best verified choice with respect to SMAPE) is presented in Table 3.

Table 3 SMAPE values of the approach with suppressed opposite correlations [%]

Method\Waste type	Paper	Plastics	Mixed MSW	Total
Suppressed terms $\boldsymbol{\varepsilon}^{-}\boldsymbol{D}\boldsymbol{Q}_{\text{spectral}}\boldsymbol{D}\boldsymbol{\varepsilon}^{+}$	17.2	8.256	5.024	10.16

Table 4 Analysis of the produced forecasts

Does the original approach outperform base forecasts?			
Yes, 58%		*No,* 42%	
Does the approach with correlation outperform the original approach?		Does the approach with correlation outperform the original approach?	
Yes, 33%	*No,* 25%	*Yes,* 22%	*No,* 20%

According to Table 3, the modified approach provides more accurate predictions than the initial system of weights. The further discussion will be focused solely on the approach (48–52). Clearly, the proposed methodology can be useful for prediction with short time HTS with noised data. However, the obtained results have not fulfilled the expectations that correlation will produce demonstrably better forecasts. The further analysis has revealed that the initial weights $\boldsymbol{D}$, require further improvement: they provide results, that are closer to the real data (compared to original predictions) only in 58% of time series. Possibly, due to imprecisions produced by the original system of weights, the correlation might bring additional noise and produce worse forecasts that are even more distant from the data. The information presented in Table 4 supports this hypothesis. In 55% of the time series the correlation has improved the original methodology, which was the main aim.

Moreover, the correlation structure helps to achieve higher precision than the original system of weights, when these weights produce better results than the base forecasts. To prove this claim, the SMAPE values for the time series, where reconciliation performed better, than the base forecasts, have been computed in Table 5.

Thus, it can be concluded, that the suggested modification produces forecasts which are demonstrably better, than the ones provided by the objective function without correlation, but only under the assumption that the base forecasts are outperformed by the original system of weights. Moreover, even these results might be slightly distorted by the strong correlation with the time series where reconciliation does not outperform

Table 5 SMAPE values of the original and suppressed correlation approaches computed for the time series, where initial reconciliation has outperformed base forecasts [%]

Method\Waste type	Paper	Plastics	Mixed MSW	Total
Without correlation	18.055	8.152	4.795	10.962
Suppressed terms $\boldsymbol{\varepsilon}^{-}\boldsymbol{D}\boldsymbol{Q}_{\text{spectral}}\boldsymbol{D}\boldsymbol{\varepsilon}^{+}$	17.262	7.951	4.363	10.483

base forecasts. Thus, one of the crucial aims of the future research is improvement of the original reconciliation methodology. From such an improvement, the approach considering correlation is expected to exceptionally benefit. It will be also beneficial to generalize the proposed approach and validate it on other available hierarchical data sets.

5 Conclusion

This article has presented the problem of the MSW generation HTS forecasting. The main aim was to satisfy the demand for more precise novel forecasting approaches, that will provide quality input for the tasks of decision-making within MSWM, while considering possible dependencies between MSW fractions and distinct administrative division levels. The correlation between MSW fractions might bring additional information about waste sorting trends present in the data, which is neglected in current state-of-the-art of MSW forecasting, where mostly only one type of waste is considered. Frequently applied methods ensuring aggregational consistency of forecasts have been thoroughly reviewed. It has been concluded that the reconciliation of the forecasts is the best option for the presented data set. Reconciliation is the current trend in HTS forecasting: it is acknowledged that reconciliation is able to improve quality of the base forecasts. The novel game-theoretic approach to the reconciliation, which considers interdependencies between the data series has been precisely defined and its theoretical properties has been studied in detail. It has been theoretically proven, that the considered correlation matrix estimate will guarantee existence and uniqueness of the resulting solution. The application of the proposed method to the MSWM data of micro-regions in the Czech Republic has demonstrated that direct application of the approach does not outperform the original approach: total SMAPE value was worse by approximately 1% compared to the 10.37% result achieved via the approach without correlation structure. However, the properties of the problem made it possible to suggest modification, where parts of correlation, corresponding to the interaction of multiplicative errors' variables of distinct type ($\boldsymbol{\varepsilon}^-$ and $\boldsymbol{\varepsilon}^+$), have been suppressed. This modification has outperformed original approach by approximately 0.2%. Though, the results were more precise, they did not demonstrate theoretically expected substantial improvement. This fact has been addressed by the hypothesis that the proposed approach is able to improve times series only when the initial system of weights outperforms base forecasts. The further analysis of the results speaks in favor of this hypothesis: it was demonstrated that in 33% of time series, where base forecasts have been outperformed by both the original approach and the proposed method with suppressed correlation, the suppressed correlation method performs better by approximately 0.5% on average. Future research should be mainly devoted to improvement of the initial weights system. It will be also beneficial to study a larger set of time series describing more waste fractions to check how it will affect the performance of the proposed method, where potential problems with computational complexity will be solved by appropriately designed decomposition.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

Ethical approval The ethical approval declaration is not applicable to the study.

Consent for publication All authors consent to the publication of this manuscript.

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